

SAMAN BAHZAD KHAN

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EDUCATION

Lahore University of Management Sciences – B.S. in Computer Science, Lahore, Pakistan Aug 2023 – *present*
Relevant Coursework: Data Structures & Algorithms, Operating Systems, AI/ML, Generative AI, Advanced GenAI, AI on edge devices, Network Centric Computing, Topics in LLMs

RESEARCH EXPERIENCE

Sem-DPO: Mitigating Semantic Inconsistency in Preference Optimization Summer 2024 – Spring 2025
Advisors: Dr. Ali Anwar, University of Minnesota · Dr. Ahmad Faraz Khan, Google

- Developed a weighted Direct Preference Optimization approach that reduces semantic drift during automated prompt optimization and established a theoretical bound on drift in preference-tuned prompt generators.
- Improved **CLIP similarity by 8–12%** and **human-preference scores by 5–9%** over standard DPO across three benchmarks and three language-model backbones; preferred in **82.2% of pairwise human evaluations**.

Impact: *Published at **ACL’26**.*

LADs: Leveraging LLMs for AI-Driven DevOps Fall 2025 – Spring 2026
Advisors: Dr. Ali Anwar, University of Minnesota · Dr. Ali R. Butt, Virginia Tech **First Author**

- Co-developed a six-agent framework for configuration, deployment, monitoring and automated repair; achieved **68% deployment success** across **180 Dask, Redis and Ray tasks** and repaired **94% of runtime failures within three iterations**.
- Adapted two existing systems as baselines and substantially redesigned the evaluation and technical presentation to better establish LADs’ gains over prior approaches.

Impact: *Submission at **MLSys’27**.*

SkillGraph: Reusable Reasoning Skills from LLM Agent Traces Summer 2026 – Present
Advisors: Dr. Tianyin Xu, University of Illinois Urbana-Champaign · Dr. Ali Anwar, University of Minnesota **First Author**

- Leading development of a GraphRAG-based skill graph for extracting and reusing diagnostic reasoning from agent trajectories; built a **database of reasoning traces** for skill extraction and retrieval on new system incidents.
- Identified shortcut behavior that corrupts learned skills and designed a **process-level critic** that evaluates reasoning quality rather than final outcomes alone.

Impact: *Submission at **MLSys’27**.*

Agent Memory: Storage and Retrieval of LLM Reasoning Traces at Scale Summer 2026 – Present
Advisor: Dr. Ali Anwar, University of Minnesota

- Analyzed **535 queries across seven models**, finding reasoning traces from **<1 KB to >71 KB per query**; developing co-location and indexing strategies for highly variable semantic reasoning workloads.

Impact: *Submission at **MLSys’27**.*

Federated Skill Sharing for LLM Agents Summer 2026 – Present
Advisors: Dr. Ali R. Butt, Virginia Tech · Dr. Ahmad Faraz Khan, Google

- Developing a federated framework where clients share reusable skill templates and generalized reasoning patterns through a global indexed library while keeping **private context and final adaptation local**.

Impact: *Submission at **MLSys’27**.*

WORK EXPERIENCE

Systems Limited, Lahore – Summer Intern Jun 2025 – Jul 2025

- Built an ERP workflow automating procurement, project tracking and reporting for a construction company using Microsoft Dynamics.

SELECTED PROJECTS

- **LLM Affordability Index (LLAI):** Developed a scenario-aware affordability framework incorporating model performance, deployment cost and regional accessibility; evaluated across five LLMs.
- **CAPS Counselling Appointment System:** Built a Firebase-backed Android scheduling platform with Google Meet integration, AI assistance and double-booking prevention.

TECHNICAL PROFICIENCY

- **Programming Languages:** Python, Java, C++, C, TypeScript, SQL
- **Tools and Libraries:** PyTorch, TensorFlow, Hugging Face, LangChain, Ollama, scikit-learn, Pandas, Git, Docker, Dask, Linux

ADDITIONAL EXPERIENCES & SERVICES

- Organized Girls’ Coding Camp, Women in Computing Hackathon, Grad App Wars and PsiFi through LUMS Women in Computing and SPADES.
- Reviewed papers for IEEE and ACM conferences; conducted Lahore air-quality analysis using PM_{2.5}, GIS mapping and Python.